

# Darshil Kalyani

Montreal, QC | 263-882-2444 | [d\\_kalyan@live.concordia.ca](mailto:d_kalyan@live.concordia.ca) | [GitHub](#) | [LinkedIn](#)

## EDUCATION

### Concordia University

*Master of Applied Computer Science*

Sept. 2025 – Present

*Montreal, QC*

### Indus University

*Bachelor of Technology in Computer Science and Engineering*

2021 – 2025

*Ahmedabad, India*

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, SQL, PHP, HTML, CSS

**Web & Databases:** React, Next.js, Express.js, Django, Play, REST APIs, MySQL, MongoDB, SQLite

**Data & Analytics:** Pandas, NumPy, SciPy, Matplotlib, Plotly, Excel, Power BI (bootcamp)

**AI & Search:** scikit-learn, PyTorch, Hugging Face, Sentence Transformers, Elasticsearch, LLM APIs

**Testing & Tools:** JUnit, Mockito, AssertJ, JaCoCo, Git, Jira, Agile, Jupyter, AI coding tools

**Cloud & Automation:** AWS S3, BullMQ, OpenTelemetry, PyAutoGUI, CI/CD, GitHub Actions, Jenkins

## EXPERIENCE

### Physical Research Laboratory (PRL)

*Software Development Intern*

Jan. 2025 – Apr. 2025

*Ahmedabad, India*

- Developed **Python** tools to automate telescope observation, CCD imaging, and calibration workflows, reducing manual operator effort and standardizing procedures for astronomers.
- Built Tkinter interfaces with start/stop controls and threaded execution; validated filters, exposure times, frame counts, and observation sequences before running automated workflows.
- Processed star catalogs using **Pandas, NumPy, SciPy, and Astropy**; filtered coordinates, magnitudes, and angular separation to identify suitable reference stars for telescope observations.
- Collaborated with a scientist and fellow intern through technical reviews and incremental **testing**; validated software remotely and on live telescope hardware at Mt. Abu Observatory.

### Plumscope Inc.

*Frontend Web Development Intern*

May 2023 – July 2023

*Ahmedabad, India*

- Developed two production webpages using HTML, CSS, and **JavaScript**; implemented responsive layouts and contact-form validation, supporting MySQL-backed form data persistence.
- Participated in **requirements discussions**, functional and cross-browser testing, review, and production deployment within a three-person team.

## PROJECTS

### NotiLytics | *Java, Play Framework, NewsAPI, JUnit, Mockito, JaCoCo*

- Built a news analytics **web application** integrating **REST APIs** for sentiment scoring and readability metrics; separated controllers and services using MVC and dependency injection.
- Implemented asynchronous API calls with `CompletableFuture`, handled timeouts and malformed responses, and isolated session search histories with `ConcurrentHashMap`.
- Wrote controller and model **unit tests** with JUnit, Mockito, and AssertJ; mocked external APIs for deterministic failure-path tests and used JaCoCo to identify untested branches.

### DocuMind | *React, Next.js, TypeScript, Express.js, MongoDB, AWS S3*

- Built a **full-stack application** with **React/Next.js** and a TypeScript/Express.js backend; implemented conversation management, MongoDB persistence, and streamed responses through LLM APIs.
- Integrated multiple AI providers through the Vercel AI SDK; implemented JWT/OAuth authentication, validated **AWS S3** uploads, BullMQ background processing, and OpenTelemetry tracing.

### BioPredict | *Python, Django, scikit-learn, Pandas, NumPy, SQLite*

- Prepared heart disease, diabetes, and lung cancer datasets through **missing-value handling**, normalization, and feature selection; trained classification models using scikit-learn.
- Compared Random Forest, SVM, Decision Tree, and Logistic Regression using **cross-validation**, accuracy, precision, recall, and F1; integrated serialized models into Django forms and prediction views.